



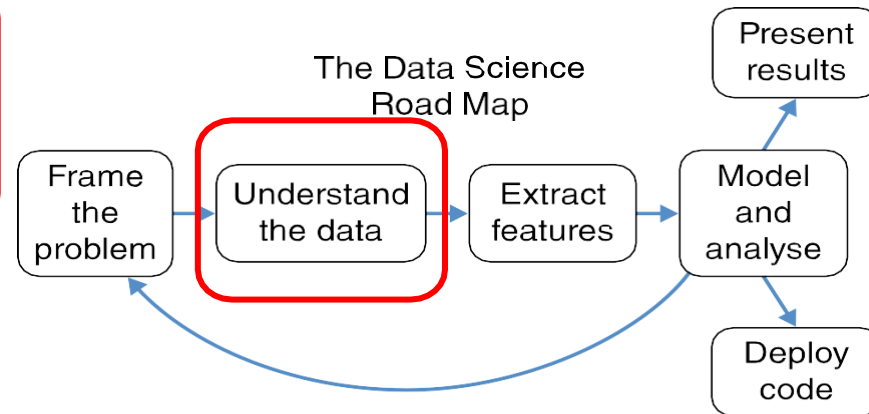
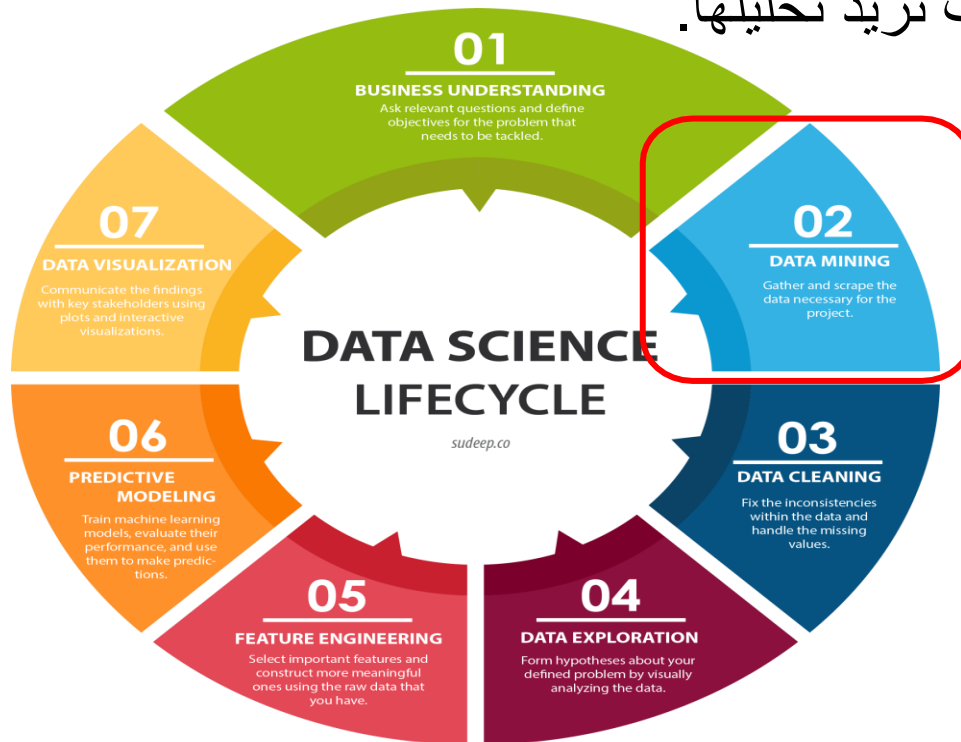
DATA SCIENCE

DATA UNDERSTANDING AND PREPARATION

DATA UNDERSTANDING AND PREPARATION

- Before you can dive deeply, you must better understand what is in your data, which will inform how you want to analyse it.

قبل أن تتمكن من الغوص بعمق ، يجب أن تفهم ما هو أفضل موجودة في بياناتك ، والتي ستعلم كيف تريد تحليلها.



UNDERSTAND THE DATA

BASIC QUESTIONS

Once you have access to the data you'll be using, **it's good to have a battery of standard questions** that you always ask about it.

- This is a good way to hit the ground running with your analyses, rather than risk analysis paralysis.
- It is also a good safeguard to identify problems with the data as quickly as possible.

أسئلة أساسية

بمجرد أن تتمكن من الوصول إلى البيانات التي ستستخدمها ، من الجيد أن يكون لديك مجموعة من الأسئلة القياسية التي تطرحها دائماً عنها.

هذه طريقة جيدة للركض مع تحليلات بدلاً من شلل تحليل المخاطر.

إنها أيضاً ضمانة جيدة لتحديد المشكلات المتعلقة بالبيانات بأسرع ما يمكن.

UNDERSTAND THE DATA

BASIC QUESTIONS

The **most important** question to ask about the data is **whether it can solve the business problem that you are trying to tackle or not?**

- If not, then you might need to look into additional sources of data or modify the work that you are planning.

أسئلة أساسية

إن أهم سؤال يجب طرحه حول البيانات هو ما إذا كان بإمكانها حل مشكلة العمل التي تحاول معالجتها أم لا؟

إذا لم يكن الأمر كذلك ، فقد تحتاج إلى البحث في مصادر إضافية لبيانات أو تعديل العمل الذي تخطط له.

UNDERSTAND THE DATA

BASIC QUESTIONS

A few good **generic questions** to ask are as follows:

- How big is the dataset?
- Is this the entire dataset?
- Is this data representative enough?
 - E.g.: maybe data was only collected for a subset of users.
- Are there likely to be gross outliers or extraordinary sources of noise?
 - E.g.: 99% of the traffic from a web server might be a single denial-of-service attack.
- Might there be artificial data inserted into the dataset?
 - This happens a lot in industrial settings.

أسئلة أساسية
بعض الأسئلة العامة الجيدة التي يجب طرحها هي كما يلي:
ما هو حجم مجموعة البيانات؟
هل هذه مجموعة البيانات بأكملها؟
هل هذه البيانات ممثل كاف؟
على سبيل المثال: ربما تم جمع البيانات فقط لمجموعة فرعية من المستخدمين.

هل من المحتمل أن تكون هناك قيم متطرفة أو مصادر غير عادية للضوضاء؟
على سبيل المثال: 99% من حركة المرور من خادم الويب قد تكون هجومًا واحدًا لرفض الخدمة.

هل يمكن إدخال بيانات اصطناعية في مجموعة البيانات؟
يحدث هذا كثيرًا في البيئات الصناعية.

UNDERSTAND THE DATA

BASIC QUESTIONS

A few good **generic questions** to ask are as follows:

- Are there any fields that are unique identifiers?
 - These are the fields you might use for joining between datasets, etc.
- Are the supposedly unique identifiers actually unique?
 - What does it mean if they aren't?
- If there are two datasets A and B that need to be joined, **what does it mean if something in A doesn't matching anything in B?**
- When data entries are blank, where does that come from?
- How common are blank entries?

أسئلة أساسية

بعض الأسئلة العامة الجيدة التي يجب طرحها هي كما يلي:

هل هناك أي حقول تعتبر معرفات فريدة؟

هذه هي الحقول التي قد تستخدمها للربط بين مجموعات البيانات ، وما إلى ذلك.

هل المعرفات الفريدة المفترضة فريدة بالفعل؟

ماذا يعني إذا لم يكونوا كذلك؟

إذا كانت هناك مجموعتا بيانات "أ" و "ب" يلزم ضمهما ، فماذا يعني إذا كان هناك شيء في "أ" لا يتطابق مع

أي شيء في "ب"؟

عندما تكون إدخالات البيانات فارغة ، من أين يأتي ذلك؟

ما مدى شيوع الإدخالات الفارغة؟

UNDERSTAND THE DATA

BASIC QUESTIONS

A few good **generic questions** to ask are as follows:

- What data is available for the task?
- Is this data relevant?
- Is additional relevant data available?
- How much historical data is available?
- Who is the data expert and Domain Expert?

أسئلة أساسية

بعض الأسئلة العامة الجيدة التي يجب طرحها هي كما يلي:

ما هي البيانات المتاحة للمهمة؟

هل هذه البيانات ذات صلة؟

هل البيانات الإضافية ذات الصلة متاحة؟

كم البيانات التاريخية المتاحة؟

من هو خبير البيانات وخبير المجال؟

UNDERSTAND THE DATA

DATA TYPES

- Data can be broadly classified **into four types**:

1. **Structured Data:**

- Have a predefined model, which organizes data into a form that is relatively easy to store, process, retrieve and manage
 - E.g., relational data

2. **Unstructured Data:**

- Opposite of structured data
 - E.g., Flat binary files containing text, video or audio

- **Note:** data is not completely devoid of a structure

- E.g., an audio file may still have an encoding structure and some metadata associated with it.

أنواع البيانات

يمكن تصنيف البيانات على نطاق واسع إلى أربعة أنواع:
البيانات المنظمة:

لديك نموذج محدد مسبقاً ، ينظم البيانات في نموذج يسهل نسبياً تخزينه ومعالجته واستردادته وإدارته
على سبيل المثال ، البيانات العلائقية
البيانات غير المنظمة:

عكس البيانات المنظمة

على سبيل المثال ، ملفات ثنائية مسطحة تحتوي على نص أو فيديو أو صوت
ملاحظة: البيانات ليست خالية تماماً من هيكل

على سبيل المثال ، قد يظل ملف الصوت بهيكل تشفير و
بعض البيانات الوصفية المرتبطة به.

UNDERSTAND THE DATA

DATA TYPES

▪ Data can be broadly classified into four types:

3. Dynamic Data:

- Data that changes relatively frequently
 - E.g., office documents and transactional entries in a financial database

4. Static Data:

- Opposite of dynamic data
 - E.g., Medical imaging data from MRI or CT scans

أنواع البيانات

يمكن تصنيف البيانات على نطاق واسع إلى أربعة أنواع:

البيانات الديناميكية:

البيانات التي تتغير بشكل متكرر نسبيًا

على سبيل المثال ، المستندات المكتوبة وإدخالات المعاملات في قاعدة بيانات مالية

بيانات ثابتة:

عكس البيانات الديناميكية

على سبيل المثال ، بيانات التصوير الطبي من التصوير بالرنين المغناطيسي أو التصوير

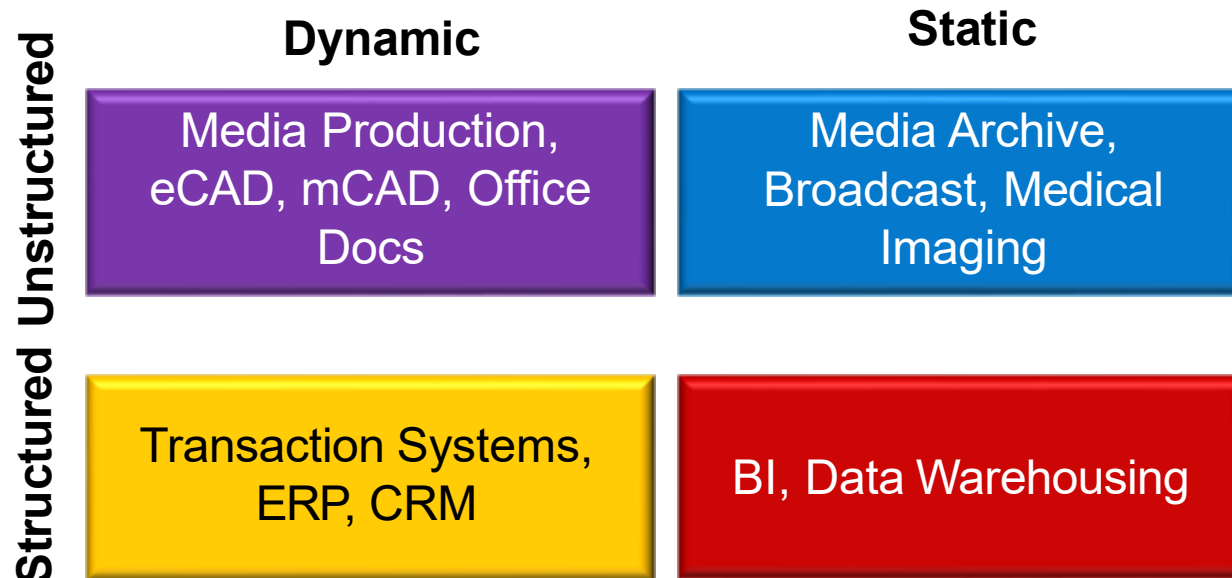
المقطعي المحوسب

UNDERSTAND THE DATA

DATA TYPES

▪ Segmenting data into one of the categories can help in designing and developing a pertaining solution.

▪ يمكن أن يساعد تقسيم البيانات إلى إحدى الفئات في تصميم وتطوير حل ذي صلة.



UNDERSTAND THE DATA

DATA SETS

Record

- Relational records
- Data matrix,
 - e.g., numerical matrix, crosstables
- Document data:
 - text documents: term-frequency vector
- Transaction data

Graph and network

- World Wide Web
- Social or information networks
- Molecular Structures

Ordered

- Video data
 - sequence of images
- Temporal data
 - time-series
- Sequential Data
 - transaction sequences
- Genetic sequencedata

Spatial, image and multimedia:

- Spatial data
 - maps
- Image data
- Video data

	team	cdach	pl a	ball	score	game	i w	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

<i>TID</i>	<i>Items</i>
1	Bread,Coke,Milk
2	Beer,Bread
3	Beer,Coke,Diaper,Milk
4	Beer,Bread,Diaper,Milk
5	Coke,Diaper,Milk

UNDERSTAND THE DATA

DATA SETS: DATA OBJECTS

Data sets are made up of **data objects**.

A **data object** represents **an entity**.

- Examples:
 - *sales database*: customers, store items, sales
 - *medical database*: patients, treatments
 - *university database*: students, professors, courses
- Also it is called **sample** , **example**, **instance**, **data point**, and **tuple**.

Data objects **are described by attributes**.

- In database
 - rows -> data objects
 - columns -> attributes.

UNDERSTAND THE DATA

DATA SETS: ATTRIBUTES

Attribute is a **data field**, representing a characteristic **or** feature of a data object.

- E.g., customer_ID, name, address
- Also it is called as **dimensions**, **features**, and **variables**.

Attribute's Types:

- Nominal
- Binary
- Ordinal
- Numeric:
 - Quantitative
 - Interval-scaled
 - Ratio-scaled

UNDERSTAND THE DATA

DATA SETS: ATTRIBUTE'S TYPES

Nominal:

- categories, states, or “names of things”
 - color = {black, blue, brown, grey, red, white}
 - marital status, occupation, ID numbers, zip codes

Binary

- Nominal attribute with only 2 states (0 and 1)
- Symmetric binary: both outcomes equally important
 - e.g., gender
- Asymmetric binary: outcomes not equally important.
 - e.g., medical test (positive vs. negative)
 - Convention: assign 1 to most important outcome

UNDERSTAND THE DATA

DATA SETS: ATTRIBUTE'S TYPES

Ordinal

- Values have a meaningful order (ranking) but magnitude between successive values is not known.
 - Size = {small, medium, large},
 - Grades = {Excellent, Very Good, Good, Weak}
 - Army Rankings.

Numeric

- The attribute's values are usually represented by a numeric values and it can be classified as Quantitative, Interval-Scaled or Ratio-Scaled.

UNDERSTAND THE DATA

DATA SETS: ATTRIBUTE'S TYPES (NUMERIC)

Quantitative

- The values usually represent a quantity like age, weight, salary ...etc.
 - integer
 - real-valued

Interval - Scaled

- Measured on a scale of equal-sized units
- Values have order
 - E.g., temperature in C° or F°, calendar dates
- No true zero-point

Ratio - Scaled

- Inherent zero-point
- We can speak of values as being an order of magnitude larger than the unit of measurement (10 K° is twice as high as 5 K°).
 - e.g., temperature in Kelvin, length, counts, monetary quantities

UNDERSTAND THE DATA: DATASET

DISCRETE VS. CONTINUOUS ATTRIBUTES

Discrete Attribute

- Has only a finite or countably infinite set of values
 - E.g., zip codes, profession, or the set of words in a collection of documents
- Sometimes, represented as integer variables
- Note: Binary attributes are a special case of discrete attributes

Continuous Attribute

- Has real numbers as attribute values
 - E.g., temperature, height, or weight
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables

UNDERSTAND THE DATA

EXPLORATORY ANALYSIS

Once you have the data digested into a usable format, the next step is exploratory analysis.

This basically means poking around in the data, visualizing it in lots of different ways, trying out different ways to transform it, and seeing what there is to see.

This stage is very creative, and it's a great place to let your curiosity run a little wild.

- but keep things simple and intuitive.

UNDERSTAND THE DATA

EXPLORATORY ANALYSIS

There are two things that you typically get out of exploratory analysis:

1) You develop an **intuitive feel** for the data, including what the **salient patterns** look like visually.

- This is especially important if you're going to be working with similar data a lot in the future.

2) You get a list of **concrete hypotheses** about what's going on in the data.

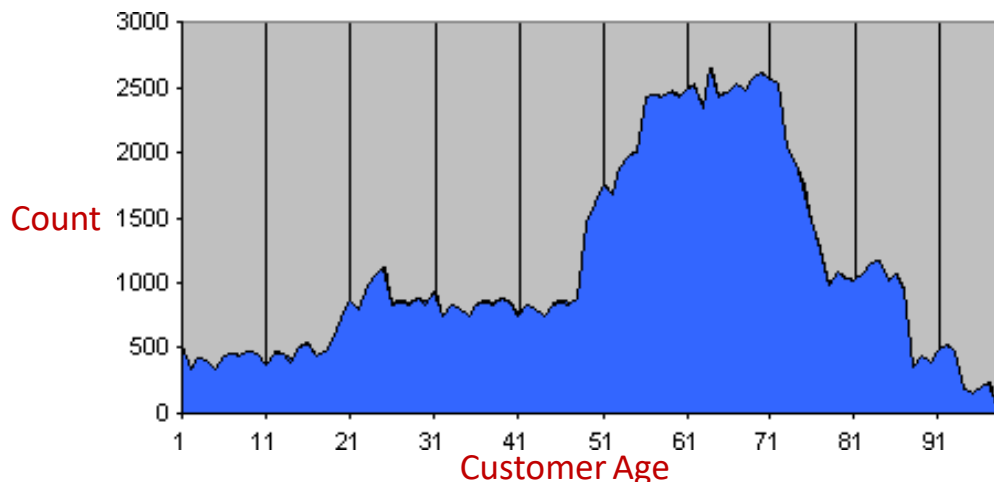
- Oftentimes, a hypothesis will be motivated by a compelling graphic that you generated: a snapshot of a time series that shows an unmistakable pattern, a scatterplot demonstrating that two variables are related to each other, or a histogram that is clearly bimodal.

UNDERSTAND THE DATA

EXPLORATORY ANALYSIS

A common misconception is that data scientists don't need visualizations.

- This attitude is not only inaccurate: it is very dangerous. Most machine learning algorithms are not inherently visual, but it is very easy to misinterpret their outputs if you look only at the numbers; there is no substitute for the human eye when it comes to making intuitive sense of things.



DATASET GATHERING

Datasets Sites:

Some examples from

- **Kaggle**
 - <https://www.kaggle.com/datasets>
- **UCI MachineLearning Repository**
 - <http://mlr.cs.umass.edu/ml/>
- **Data Quest**
 - **Places to Find Free Data Sets for Data Science Projects**
 - <https://www.dataquest.io/blog/free-datasets-for-projects/>

DATA PREPARATION (WRANGLING)

Data wrangling is **the process of getting the data from its raw format into something suitable for more conventional analytics.**

- This typically means creating a software pipeline that gets the data out of wherever it is stored, does any cleaning or filtering necessary, and puts it into a regular format.

Data wrangling **is the main area where data scientists need skills that a traditional statistician or analyst doesn't have.**

- Preparation, Preprocessing, Mugging, Wrangling, Transformation

DATA PREPARATION

PYTHON'S TECHNICAL LIBRARIES

Python was designed mostly as a tool for software engineers, but there is an excellent suite of libraries available that make it a first-class environment for technical computing, competing with the likes of MATLAB® and R.

The main ones that will be used in data preparation are:

- *NumPy*: This is a library for dealing with numerical arrays in ways that are fast and memory efficient.
 - but it's low level for a user.
- *Pandas*: This is the big one for you to know. It stores and operates on data in data frames, very efficiently and with a smooth, intuitive API.
 - Under the hood, Pandas operates on NumPy arrays.

USING PANDAS DATAFRAME

Generate a new dataset by using *Pandas DataFrame*

```
import pandas as pd
df = pd.DataFrame(
    { "name": [ "Ahmad", " Beesan", " Hani ", " Rami ", " Reema" ],
      "age": [24, 53, 23, 24, 33],
      "gender": ["M","F", "M", "M", "F"],
      "job":["technician", "other", "writer", "technician", "other"],
      "zip_Code" : [85711, 94302, 32076, 56321, 14223]},
    index = [1, 2, 3, 4, 5]
)

print (df)
```

	name	age	gender	job	zip_Code
1	Ahmad	24	M	technician	85711
2	Beesan	53	F	other	94302
3	Hani	23	M	writer	32076
4	Rami	24	M	technician	56321
4	Reema	33	F	other	14223

USING PANDAS READ_CSV

```
#load existing dataset stored as csv file
dataset = pd.read_csv("/Users/itc3/Documents/iris.csv")

# print(dataset.head()) # first five rows
# print(dataset.tail()) # last five rows
print(dataset.dtypes)    # show the dataset attribute data types

## get the dataset size and dimension
print("Shape ", dataset.shape)
print("Dimensions ", dataset.ndim)

## get columns' titles
titles = dataset.columns
print(titles)

#describing a certain column with some basic statistics
print(dataset['sepal.length'].describe())
```

WHY PREPROCESS THE DATA?

The real-world database typically used in data science may have *millions of records* and thousands of variables.

- It is *noisy* and has *missing* and *inconsistent* values.

Data quality is a key issue with data Science

Data preparation is a necessary step for serious, effective, real-world data science

WHY DO WE PREPARE THE DATA?

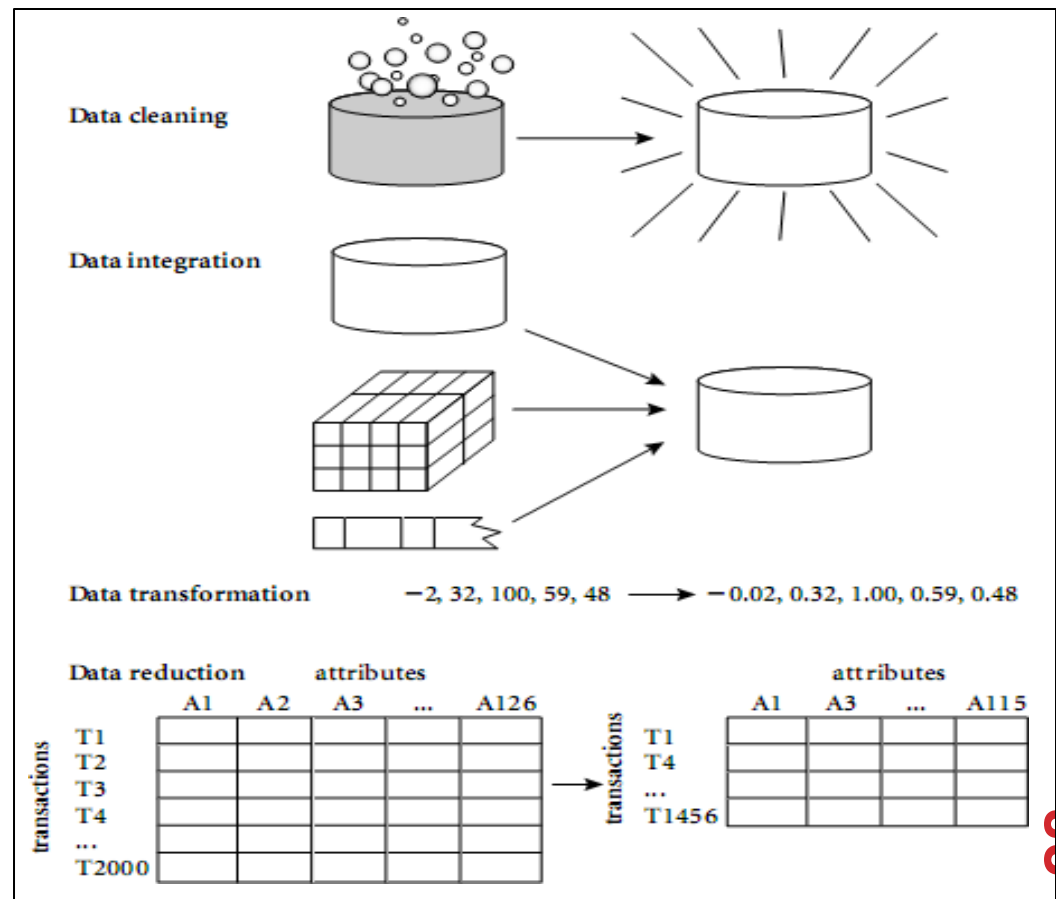
Measures for data quality: A multidimensional view

- *Accuracy*: correct or wrong, accurate or not
 - garbage in => garbage out
- *Completeness*: not recorded, unavailable, ...
- *Consistency*: some modified but some not
- *Timeliness*: timely update?
- *Believability*: how trustable the data are correct?
- *Interpretability*: how easily the data can be understood?

DATA PREPARATION ...

Data Preparation estimated to take *70-80%* of the time and effort because it always done manually.

Data Preparation Tasks



DATA PREPROCESSING TASKS

Data Cleaning

- Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies

Data Integration

- Integration of multiple databases, data cubes, or files

Data Transformation

- Discretization
- Normalization
- Concept hierarchy generation

Data Reduction

- Dimensionality reduction
- Numerosity reduction
- Data compression

DATA CLEANING

Data in the real world **is dirty**, lots of potentially **incorrect data**:

- ***incomplete***: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., Occupation=" " (missing data)
- ***noisy***: containing noise, errors, or outliers
 - e.g., Salary="-1200" (an error)
- ***inconsistent***: containing discrepancies in codes or names, e.g.,
 - Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - discrepancy between duplicate records

DATA CLEANING

INCOMPLETE (MISSING) DATA

Data **is not always available**

- E.g., many tuples have no recorded value for several attributes, such as customer income in sales data

Missing data may be due to:

- Equipment malfunction
- inconsistent with other recorded data and thus deleted
- data not entered due to misunderstanding
- certain data may not be considered important at the time of entry

DATA CLEANING:

HOW TO HANDLE MISSING DATA?

We have two solutions to handle the missing data

- **Ignore the tuple:**
 - Usually done when class label is missing, within a classification tasks.
 - Not effective when the % of missing values per attribute varies considerably
- **Fill in the missing value:**
 - **manually**
 - Tedious (boring)
 - Infeasible
 - **automatically**
 - A global constant : e.g., “unknown”, a new class, “?! ”
 - The attribute mean
 - the attribute mean for all samples belonging to the same class: smarter
 - The most probable value:
 - inference-based such as Bayesian formula or decision tree

DATA CLEANING:

HANDLING THE MISSING DATA - PYTHON

Ignore the tuple

```
# drop certain row that has null values  
data.drop([30], inplace=True, axis=0)
```

```
# drop rows with null value in a certain column  
data = data[pd.notnull(data['IrisType'])]
```

```
# drop all rows with missing values  
data = data.dropna()
```

DATA CLEANING:

HANDLING THE MISSING DATA - PYTHON

Fill the missing manually

```
# fill the missing value manually  
data.iloc[9,2] = 4.8
```

Fill the missing automatically

```
# fill the missing values in attribute automatically by mean,  
# median, max, min or even by a constant value  
data['sepal.length'] =  
data['sepal.length'].fillna(data['sepal.length'].mean())
```

DATA CLEANING: NOISY DATA

Noise: random error or variance in a measured variable

Incorrect attribute values may be due to:

- faulty data collection instruments
 - data entry problems
 - data transmission problems
 - technology limitation
 - inconsistency in naming convention
-
- duplicate records is another data problems which require data cleaning

DATA CLEANING:

HOW TO HANDLE NOISY DATA?

We have different solutions to handle the noisy data

- **Binning**
 - first sort data and partition into equal-frequency or equal-depth bins
 - then one can smooth by bin mean, median or boundaries
- **Regression**
 - smooth by fitting the data into regression functions
- **Clustering**
 - detect and remove outliers
- **Combined computer and human inspection**
 - detect suspicious values and check by human
 - e.g., deal with possible outliers

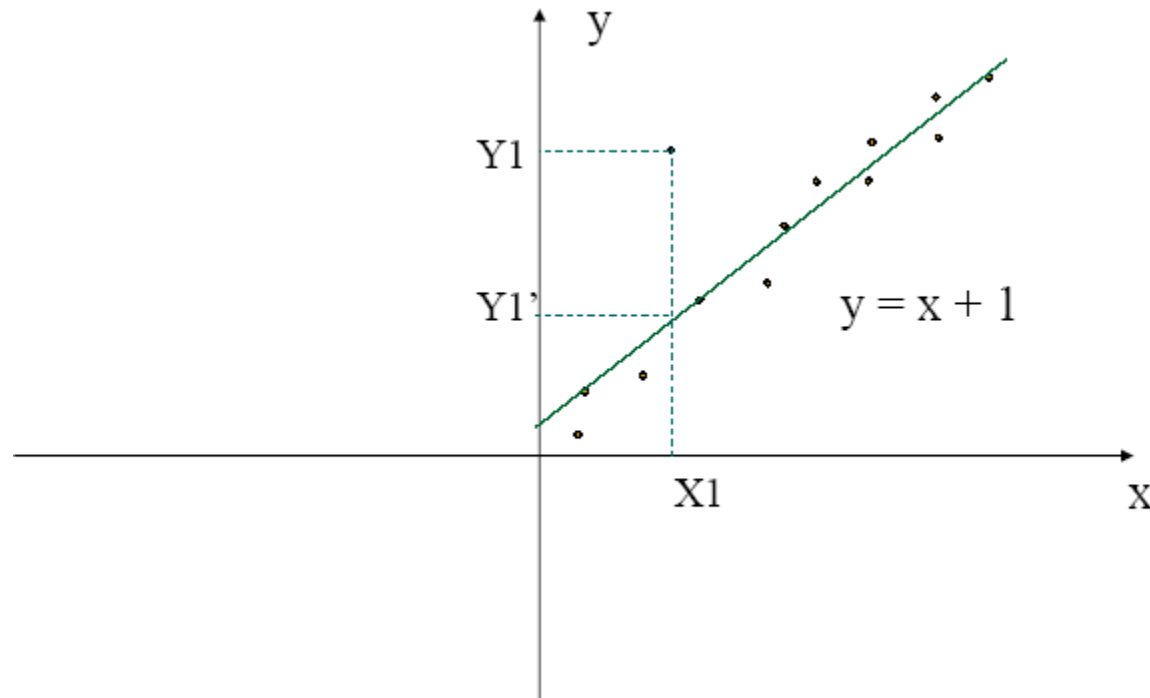
DATA CLEANING:

HANDLING NOISY DATA - BINNING

Sorted data for price : 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

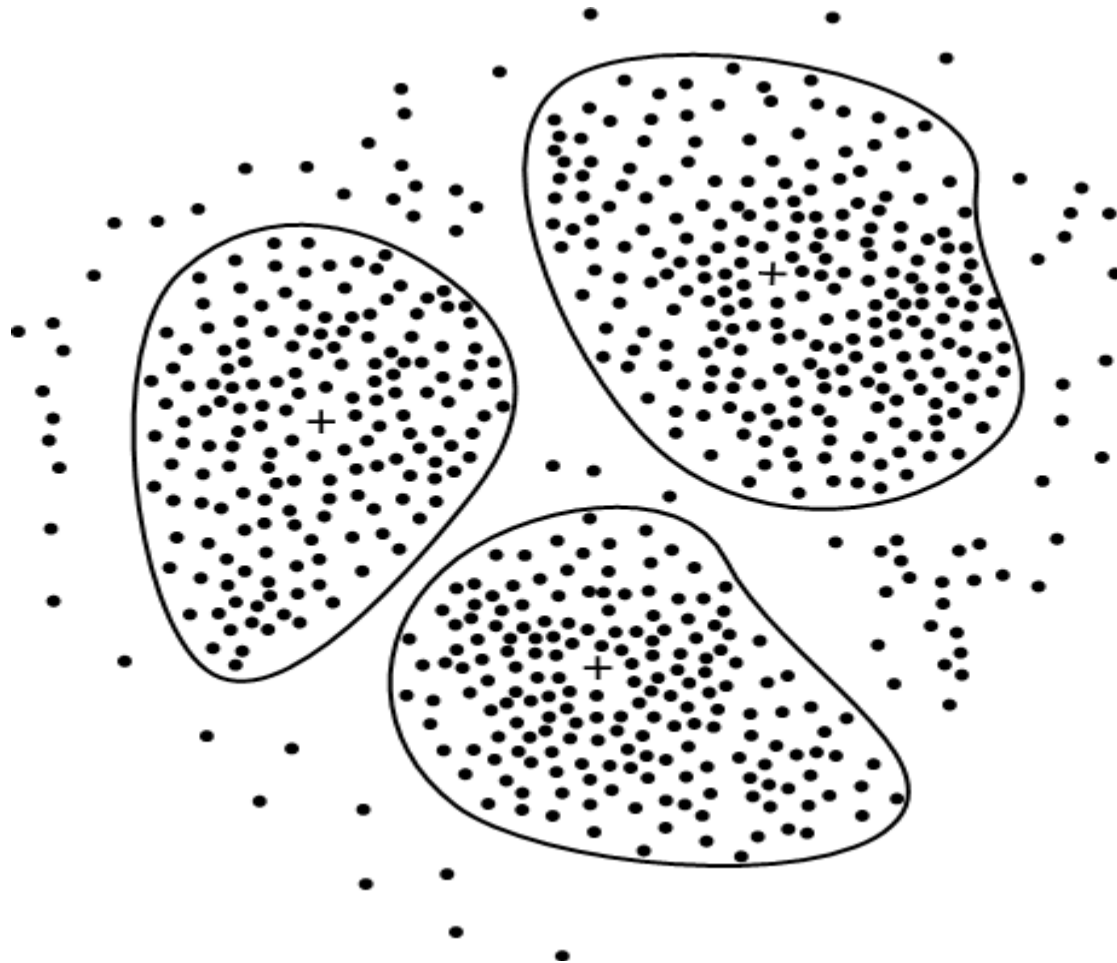
- Partition into equal-frequency bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
- Smoothing by bin means:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
- Smoothing by bin boundaries:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34

DATA CLEANING: HANDLING NOISY DATA - REGRESSION



DATA CLEANING:

HANDLING NOISY DATA - CLUSTERING



DATA CLEANING: INCONSISTENT DATA

Inconsistent Data with our models should be handled.

Common sense can be used to detect such kind of inconsistency.

- The same name occurring differently in an application
 - Different names can appear to be the same (Dennis vs Denis)

Inappropriate values

- Males being pregnant, or
- having an negative age or salary

DATA CLEANING: INCONSISTENT DATA

Data inconsistencies may be **corrected manually** using external references.

- errors made at data entry level may be corrected by **performing a papertrace**.

Example

- A particular bank's database had about 5% of it's customers born on 01/01/2000, which is usually the default value for the birthday attribute.
 - The date value will be corrected according the paper documents.

DATA CLEANING: INCONSISTENT DATA

Data inconsistency (discrepancy) detection

- Use **metadata**
 - e.g., domain, range, dependency, distribution
- Check **field overloading**
- Check **uniqueness rule**, consecutive rule and null rule
- Use **commercial tools**
 - **Data scrubbing**: use simple domain knowledge
 - e.g., postal code, spell-check) to detect errors and make corrections
 - **Data auditing**: by analyzing data to discover rules and relationship to detect violators
 - e.g., correlation and clustering to find outliers

DATA INTEGRATION

Data Integration combines data from multiple sources into a coherent datastore.

- Sources may include **multiple databases**, or **flat files**.

Why Data Integration?

- Increasingly: data science projects require data from more than one data source
- Data is stored in many systems across enterprise and outside the enterprise

DATA INTEGRATION

The source of data fall into two categories:

- **Internal sources** that are generated through enterprise activities
 - such as databases, historical data, Web sites and warehouses.
- **External sources** such as credit bureaus, phone companies and demographical information.

DATA INTEGRATION

Many problems can be occurred during Data Integration:

- **Entity Identification Problem:**
 - Identifying the real world entities in different data sources may different.
- **Data Value Conflicts (inconsistent)**
 - For the same real world entity, attribute values from different sources are different
 - this may caused by using different representation or scales
- **Redundant Data Features**
 - **Object identification:** The same attribute or object may have different names in different databases
 - **Derived data:** One attribute may be a “derived” attribute in another table,
 - e.g., annual salary

DATA TRANSFORMATION

The data are *transformed/consolidated* into forms appropriate for analysis or mining.

Data transformation methods:

- *Smoothing*: remove noise from data.
- *Aggregation*: summarization
 - e.g. daily sales aggregate into monthly or annual sales.
- *Generalization*: data are replaced by higher-level concepts through the use of concept hierarchies
 - e.g. replace city by the country.
- *Normalization*: attribute data are scaled so as to fall within a small specified range
 - e.g. temp 0.0 – 1.0 or -1.0 - 1
- *Attribute construction*: New attributes are constructed and added from the given set of attributes to help the data analysis process

DATA TRANSFORMATION

NORMALIZATION

Normalization: Scaled to fall within a smaller specified range.

- min-max normalization

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- z-score normalization

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

- decimal scaling normalization

$$v' = \frac{v}{10^j}$$

DATA TRANSFORMATION

MIN-MAX NORMALIZATION

It performs a **linear transformation** on the original data **depending on** the maximum and minimum values of the attribute and the new range values.

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

DATA TRANSFORMATION

MIN-MAX NORMALIZATION

Example:

- Suppose that the minimum and maximum values for the *income* attribute are \$12,000 and \$98,000, respectively.
- We would like to map *income* to the range [0.0;1.0].
- The value of \$73,600 for *income* will be 0.716 as follow

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

DATA TRANSFORMATION

Z-SCORE NORMALIZATION

It is known as **zero-mean** normalization

The values for an attribute A, are normalized based on the **mean** (μ), (A), and **standard deviation** (σ) of A.

$$v' = \frac{\mu - v}{\sigma}$$

- where:

- $\mu = \overline{X}$

- $\sigma = \sqrt{\frac{(X - \mu)^2}{N - 1}}$

DATA TRANSFORMATION

Z-SCORE NORMALIZATION

Example:

- Suppose that *the mean* and *standard deviation* of the values for the attribute income are *\$54,000* and *\$16,000*, *respectively*.
- The value of *\$73,600* for income will be *1.225* as follow

$$\frac{73,600 - 54,000}{16,000} = 1.225$$

DATA TRANSFORMATION

DECIMAL-SCALING NORMALIZATION

Normalizes by moving the decimal point of values of attribute A.

- The number of decimal points moved depends on the maximum absolute value of A.

$$v' = \frac{v}{10^k}$$

- where k is the smallest integer such that $\text{Max}(|v'|) < 1$.

DATA TRANSFORMATION

DECIMAL-SCALING NORMALIZATION

Example:

- Suppose that records values of attribute A range from -986 to 917.
- The **maximum absolute** value of A is **986**.
- To normalize, we therefore divide each value by 1000 ($k = 3$)
- so that **-986** normalizes to **-0.986**

DATA TRANSFORMATION

DATA REDUCTION

Data Reduction:

- Obtain a reduced representation of the data set that is **much smaller** in volume but yet **produces the same** (or almost the same) **analytical results**

Why data reduction?

- A database/data warehouse may store **Terabytes of data**. **Complex data analysis** may **takes a very long time** to run on the completed data set.
 - i.e. to **reduce the analysis time**.

DATA TRANSFORMATION

DATA REDUCTION

The choice of data **representation**, **selection**, **reduction**, or **transformation** of features are probably the most important issues that determine the quality of a data-science solution.

There are three basic operations in a data-reduction process:

- **Feature selection**, remove irrelevant attributes,
- **Sampling**, reduce the row counts, and
- **Discretization**, reduce the number of values in a column.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Features selection aims to choose the features that are relevant to the data-science application in order to achieve maximum performance with the minimum measurement and processing effort.

reduces the data set size by removing irrelevant or redundant attributes.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

“How can we find a „good“ subset of the original attributes?”

For n attributes, there are 2^n possible subsets.

There are two types of feature selection

- **Unsupervised:**
 - Reduce fields **without knowing** the class label or class target
- **Supervised:**
 - Select fields **with respect to** the class label

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Remove Redundant Features

- Duplicate much or all of the information contained in one or more other attributes
 - e.g., purchase price of a product and the amount of sales tax paid

Remove Irrelevant features

- Contain no information that is useful for the desired data science task.
 - e.g., students' ID is often irrelevant to the task of predicting students' GPA

Remove fields with no or with little variability

- Examine the number of distinct field values and its ratios
 - e.g. gender in the breast cancer data set

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

An exhaustive search for the optimal subset of attributes can be **too expensive**, especially as n and the number of data classes increase.

- Therefore, **heuristic methods** that explore a reduced search space are commonly used for attribute subset selection.
- The “**best**”, or “**worst**”, attributes are typically **determined by using tests of statistical significance**, which **assume that the attributes are independent of one another**.

تقليل البيانات: اختيار الميزات
يمكن أن يكون البحث الشامل عن المجموعة الفرعية المثلى للسِمات مكلفاً للغاية ، خاصة مع زيادة n وعدد فئات البيانات.
لذلك ، تُستخدم الطرق الإرشادية التي تستكشف مساحة بحث مخفضة بشكل شائع لاختيار مجموعة السِمات الفرعية.
عادةً ما يتم تحديد السِمات "الأفضل" أو "الأسوأ" باستخدام اختبارات ذات دلالة إحصائية ، والتي تفترض أن السِمات مستقلة عن بعضها البعض.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Forward selection

Initial attribute set:
 $\{A_1, A_2, A_3, A_4, A_5, A_6\}$

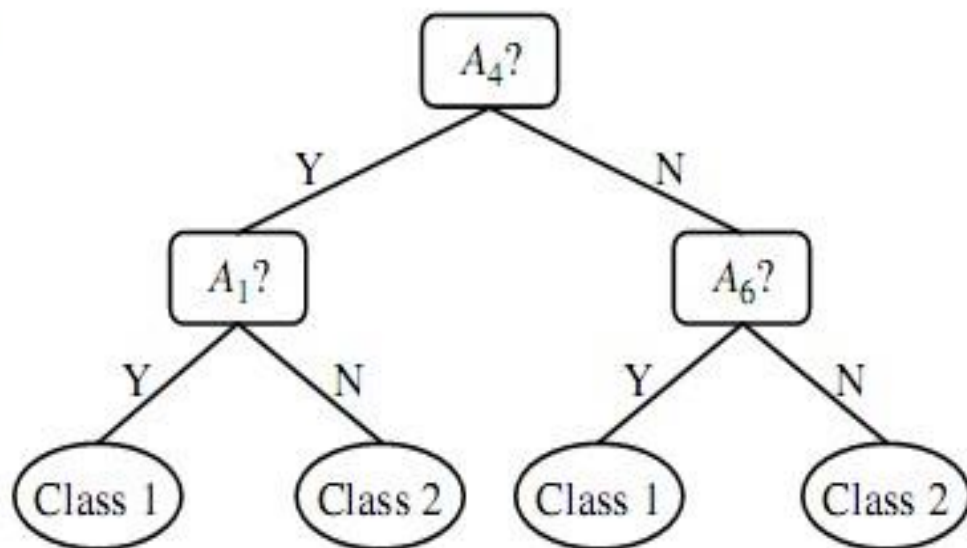
Initial reduced set:
 $\{\}$
 $\Rightarrow \{A_1\}$
 $\Rightarrow \{A_1, A_4\}$
 $\Rightarrow$ Reduced attribute set:
 $\{A_1, A_4, A_6\}$

Backward elimination

Initial attribute set:
 $\{A_1, A_2, A_3, A_4, A_5, A_6\}$
 $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$
 $\Rightarrow \{A_1, A_4, A_5, A_6\}$
 $\Rightarrow$ Reduced attribute set:
 $\{A_1, A_4, A_6\}$

Decision tree induction

Initial attribute set:
 $\{A_1, A_2, A_3, A_4, A_5, A_6\}$



$\Rightarrow$ Reduced attribute set:
 $\{A_1, A_4, A_6\}$

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Some redundancy can be detected by correlation analysis (how strongly one attribute implies the other)

Correlation analysis can be calculated by using:

- Pearson correlation coefficient test
 - Such as Pearson's product moment coefficient
 - It used for numerical attributes
- χ^2 (chi-square) test
 - It used for nominal attributes
- Spearman's Correlation
- Kendall's Tau

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Pearson correlation coefficient test

- For two numerical attributes A and B , $r_{A,B}$ can be calculated as:

$$r_{A,B} = \frac{\sum_{i=1}^N (a_i - \bar{A})(b_i - \bar{B})}{N\sigma_A\sigma_B} = \frac{\sum_{i=1}^N (a_i b_i) - N\bar{A}\bar{B}}{N\sigma_A\sigma_B}$$

- where
 - N is the number of tuples, a_i and b_i are the respective values of A and B in tuple i ,
 - $\bar{A}$ and $\bar{B}$ are the *respective mean* values of A and B
 - σ_A and σ_B are the *respective standard deviations* of A and B
 - $(a_i b_i)$ is the **sum** of the *AB cross-product*

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Pearson correlation coefficient test

- $-1 \leq r_{A;B} \leq +1$
- A and B are **positively correlated** if $r_{A;B}$ is greater than 0,
 - this means that the *values of A increase* as the *values of B increase*.
 - The **higher the value**, the **stronger the correlation**
 - The **strong correlation** indicate **that A (or B) may be removed as a redundancy**.
- A and B are **independent** if $r_{A;B}$ is equal 0
- A and B are **negatively correlated** if $r_{A;B}$ is less than 0.
 - this means that the *values of A increase* as the *values of B decrease*.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Chi-Square Test:

- For nominal data, the Chi-Square test is discovered by χ^2 test

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

- where o_{ij} is the observed frequency, the actual count, and e_{ij} is the expected frequency

The **larger** χ^2 value, the **more likely the variables are related**

The test is based on a **significance level**, with $(r - 1) \times (c - 1)$ degrees of freedom.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

Chi-Square Test Example:

- Suppose that a group of 1,500 people was surveyed. The gender of each person was noted. Each person was polled as to whether their preferred type of reading material was *fiction* or *nonfiction*. Thus, we have two attributes, **gender** and **preferred reading**.

تقليل البيانات: اختيار الميزات

Chi-Square: مثال اختبار

افترض أنه تم مسح مجموعة من 1500 شخص. لوحظ جنس كل شخص. تم استطلاع رأي كل شخص لمعرفة ما إذا كان النوع المفضل لديهم من مواد القراءة هو خيالي أم واقعي. وبالتالي ، لدينا صفتان ، الجنس والقراءة المفضلة.

	male	female	Sum (row)
fiction	250(e_{ij})	200(e_{ij})	450
non fiction	50(e_{ij})	1000(e_{ij})	1050
Sum(col.)	300	1200	1500

- The *expected frequency* e_{ij} calculated by

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{N}$$

- In a row/column, the sum of the expected frequencies must equal the total observed frequency for that row/column

• في صف / عمود ، يجب أن يساوي مجموع التكرارات المتوقعة إجمالي التردد المرصود لهذا الصف / العمود

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

	male	female	Sum (row)
fiction	250(90)	200(360)	450
non fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

With 1 degree of freedom ((2-1)X(2-1)), χ^2 value **needed to reject the hypothesis** at the 0.001 significance level is 10.827

taken from the table of upper percentage points of the χ^2 distribution, typically available from any textbook on statistics.

مع درجة واحدة من الحرية ((1-2)) ((2-1)) X، تكون قيمة 2 اللازمة لرفض الفرضية عند مستوى الأهمية 0.001 هي 10.827

مأخوذة من جدول النقاط المئوية العليا لتوزيع χ^2 ،
عادة ما تكون متاحة من أي كتاب مدرسي عن الإحصاء.

DATA TRANSFORMATION

DATA REDUCTION: FEATURES SELECTION

	male	female	Sum (row)
fiction	250(90)	200(360)	450
non fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

Since our computed value is above 10.828 , we can **reject** the hypothesis that *gender* and *preferred reading* are independent and conclude that the two attributes are (strongly) correlated for the given group of people.

نظرًا لأن القيمة المحسوبة لدينا أعلى من 10.828 ، يمكننا رفض الفرضية القائلة بأن الجنس والقراءة المفضلة مستقلتان ونستنتج أن السمتين مرتبطتان (بشدة) بمجموعة معينة من الأشخاص.

DATA TRANSFORMATION

DATA REDUCTION: SAMPLING

Sampling can be used as a data reduction technique because it allows a large data set **to be represented** by a **much smaller random sample** (or subset) of the data.

Allow a data-science algorithm to run in complexity that is potentially sub-linear to the size of the data

Key principle: Choose a representative subset of the data

- **Random sampling**, simple random sampling may have very poor performance in the presence of skew. Because there is an equal probability of selecting any particular item
- **Stratified sampling**, it is an adaptive sampling method that partitions the data set, and draw samples from each partition. The dataset almost partitioned based on class' label.

يمكن استخدام أخذ العينات كأسلوب لتقليل البيانات لأنه يسمح بتمثيل مجموعة كبيرة من البيانات بواسطة عينة عشوائية أصغر بكثير (أو مجموعة فرعية) من البيانات. السماح لخوارزمية علم البيانات بالعمل في تعقيد يحتمل أن يكون خطيًا فرعيًا لحجم البيانات المبدأ الأساسي: اختر مجموعة فرعية تمثيلية من البيانات أخذ العينات العشوائية ، قد يكون لأخذ العينات العشوائية البسيطة أداء ضعيف للغاية في وجود الانحراف. لأن هناك احتمالية متساوية لاختيار أي عنصر معين

أخذ العينات الطباقية ، هو طريقة أخذ العينات التكيفية التي تقسم مجموعة البيانات ، واستخلاص عينات من كل قسم. تم تقسيم مجموعة البيانات تقريبًا بناءً على تصنيف الفئة.

DATA TRANSFORMATION

DATA REDUCTION: DISCRETIZATION

Data discretization techniques can be used **to reduce the number of values** in specific attribute.

- i.e.: Divide the range of a continuous attribute into intervals then interval labels can then be used to replace actual data values.

Data discretization **reduces** and **simplifies** the original data and that leads to a **concise, easy-to-use, knowledge level representation** of results.

يمكن استخدام تقنيات تقدير البيانات لتقليل عدد القيم في سمة معينة.

على سبيل المثال: قسّم نطاق السمة المستمرة إلى فترات ، ثم يمكن بعد ذلك استخدام تسميات الفترات لاستبدال قيم البيانات الفعلية.

يقلل تقدير البيانات من البيانات الأصلية ويبسطها ويؤدي ذلك إلى تمثيل موجز للنتائج على مستوى المعرفة وسهل الاستخدام.

تخفيض البيانات: التمييز

Data discretization categories:

- *Unsupervised discretization*
 - Discretization without concerning with the data classes.
 - Binning by *mean* or *median*.
- *Supervised discretization*
 - Discretization with concerning with the data classes.
 - Decision tree analysis

فئات تقدير البيانات:
التكتم غير الخاضع للرقابة
التكتم دون القلق بشأن فئات البيانات.
Binning بالمتوسط أو الوسيط.
التقدير الخاضع للإشراف
التكتم فيما يتعلق بالبيانات
الطبقات.
تحليل شجرة القرار

DATA TRANSFORMATION

DATA REDUCTION: DISCRETIZATION

A simple **unsupervised discretization** method:

- consider the following data as a subset of data set

Temp	81	64	70	72	72	83	75	75	68	69	85	80	71	65
Class	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	no	no	no	no

- Sort the data according to the attribute values (temp)

Temp	64	65	68	69	70	71	72	72	75	75	80	81	83	85
Class	yes	no	yes	yes	yes	no	no	Yes	Yes	yes	no	yes	yes	no

- Discretize binning technique (median for example)

Temp	64	65	68	69	70	71	72	72	75	75	80	81	83	85
Class	yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Temp	65	65	65	70	70	70	72	72	72	80	80	80	85	85
Class	yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

DATA TRANSFORMATION

DATA REDUCTION: DISCRETIZATION

A simple **supervised discretization** method:

- consider the following data as a subset of data set

Temp	81	64	70	72	72	83	75	75	68	69	85	80	71	65
Class	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	no	no	no	no

- Sort the data according to the attribute values (temp)

Temp	64	65	68	69	70	71	72	72	75	75	80	81	83	85
Class	yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

- Discretize the data whenever the class changed (class)

Temp	64	65	68	69	70	71	72	72	75	75	80	81	83	85
Class	yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no